COMP233 - DISCRETE MATHEMATICS

Lecture 3.3

Statements with Multiple Quantifiers

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Order determines dependency.

First-Order Logic | Epp 4e, Section 3.3

Learning goals and lecture map

1 Interpret

Read nested quantifiers as ordered choices.

2 Compare

Distinguish ∀∃ from ∃∀ and same-type orders.

3 Translate

Move accurately between English and FOL.

4 Test truth

Use witnesses and counterexamples in finite models.

5 Negate

Flip every quantifier and negate the final condition.

6 Apply

Express access-control and AI knowledge rules.

Core path

Meaning | **Order** | **Translation** | **Models** | **Negation** | **Applications**

Why multiple quantifiers?

Natural language can hide two different logical structures.

“There is a person supervising every detail.”

One supervisor for all details

$$\exists p \, \forall d \;\; Supervises(p,d)$$

Each detail has some supervisor

$$\forall d \, \exists p \;\; Supervises(p,d)$$

Formal notation removes ambiguity by fixing the order of choices.

In software specifications, the two readings can define very different system requirements.

Prerequisite review: predicates and quantifiers

Predicate

A sentence containing variables that becomes true or false after values are assigned.

$$\text{Prime}(x) \qquad \text{Loves}(x,y) \qquad x < y$$

Quantifier

A symbol that states how many domain elements must satisfy a predicate.

Symbol	Meaning	To disprove
$\forall$	for every	one counterexample
$\exists$	there exists	show none works

Always state or infer the domain before judging truth.

Interpretation: who chooses first?

$$\forall x\ \exists y\ P(x,y)$$

1. Someone gives you an x.

↓

2. You choose a y that works for this x.

↓

3. Your y may depend on x.

↓

Possibly different witnesses

$$\exists y\ \forall x\ P(x,y)$$

1. You choose one y first.

↓

2. Someone may then give any x.

↓

3. The same y must work for all x.

↓

One common witness

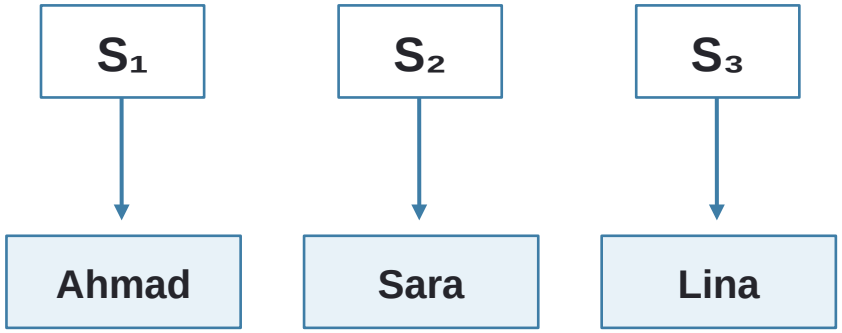
Dependency and witnesses

A witness is a value that makes an existential statement true.

Every server has a responsible person.

$$\forall s\ \exists p\ \ \text{Responsible}(p,s)$$

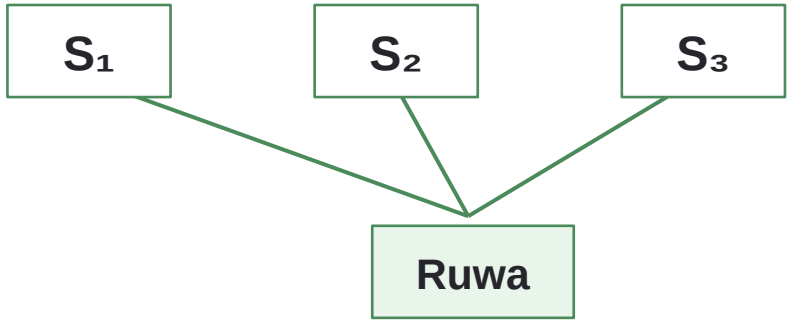
The person p may vary with the server s.



One person is responsible for every server.

$$\exists p\ \forall s\ \ \text{Responsible}(p,s)$$

One fixed person must be a common witness.



Order changes meaning - and truth

$$\forall x \in \mathbb{Z}\ \exists y \in \mathbb{Z}\ \ (x + y = 0)$$

TRUE

Given any integer x, choose y = -x.
The witness y depends on x.

Example: x = 7 → choose y = -7

$$\exists y \in \mathbb{Z}\ \forall x \in \mathbb{Z}\ \ (x + y = 0)$$

FALSE

This requires one fixed integer y that cancels every integer x.
No such common witness exists.

If y = 0, x = 1 fails. If y ≠ 0, x = 0 fails.

Mixed quantifiers are not freely interchangeable.

When can quantifiers be reordered?

Different quantifiers

$$\forall x\ \exists y\ P(x,y) \not\equiv \exists y\ \forall x\ P(x,y)$$

Reversing ∀ and ∃ usually changes the meaning and can change the truth value.

Treat mixed-order formulas as different claims.

Same quantifiers

$$\forall x\ \forall y\ P(x,y) \equiv \forall y\ \forall x\ P(x,y)$$

$$\exists x\ \exists y\ P(x,y) \equiv \exists y\ \exists x\ P(x,y)$$

Adjacent quantifiers of the same type commute.

Example: ∀x∀y (x+y=y+x)

A reliable translation workflow

1 Choose the domain

People? integers? objects in a model?

2 Define predicates

State argument order explicitly.

3 Locate quantifier words

every, each, some, one, no...

4 Preserve dependency

Read quantifiers from left to right.

5 Paraphrase back

Check that the formula says exactly what you intended.

Example **“Every student knows at least one programming language.”**

$$\forall s\,(\mathrm{Student}(s) \rightarrow \exists \ell\,(\mathrm{Language}(\ell) \wedge \mathrm{Knows}(s,\ell)))$$

Back-translation: for each student s, a language ℓ may be chosen that depends on s.

Translation matrix: the predicate Loves(x,y)

Argument order matters: x is the lover; y is the person loved.

FOL statement	Precise English
$\forall x \forall y\ \text{Loves}(x,y)$	Everyone loves everyone.
$\exists x \exists y\ \text{Loves}(x,y)$	Someone loves someone.
$\forall x \exists y\ \text{Loves}(x,y)$	Everyone loves at least one person.
$\exists y \forall x\ \text{Loves}(x,y)$	There is someone whom everyone loves.
$\forall y \exists x\ \text{Loves}(x,y)$	Everyone is loved by at least one person.
$\forall x \exists y\ (x \neq y \wedge \text{Loves}(x,y))$	Everyone loves someone other than themselves.

Multiple quantifiers with negated predicates

$\exists x \exists y \neg Loves(x,y)$	At least one person does not love at least one person.
$\forall x \forall y \neg Loves(x,y)$	Nobody loves anyone.
$\exists x \forall y \neg Loves(x,y)$	There is a person who loves nobody.
$\forall x \exists y \neg Loves(y,x)$	Every person is not loved by at least one person.

Do not weaken ∀ y ¬Loves(x,y) to “does not love everyone.” It means “loves no one.”

Evaluating truth in a finite model

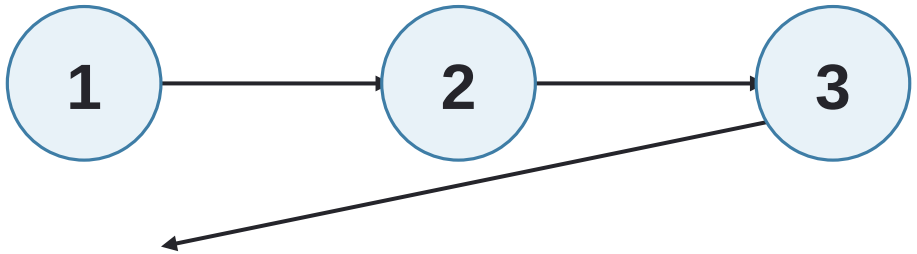
∀ outermost — Check every possible outer value. One failure disproves the statement.

∃ outermost — Find one outer value that works. If none works, the statement is false.

Inner ∃ — Provide a witness after the earlier choices are fixed.

Inner ∀ — Verify the same earlier witness survives every challenge.

Mini-model



$\forall x \exists y\ R(x,y)$ is true

Witnesses: $1 \to 2$, $2 \to 3$, $3 \to 1$.

$\exists y \forall x\ R(x,y)$ is false

Cafeteria model: the data

Predicates: Chosen(s,i) = student s chose item i; ItemIn(i,z) = item i belongs to station z.

Station / item	Uta	Tim	Yuen
SALADS			
green salad	●		
fruit salad		●	
MAIN COURSES			
spaghetti	●		●
fish		●	●
DESSERTS			
pie	●	●	●
cake		●	
BEVERAGES			
milk	●	●	
soda			●
coffee		●	

Immediate observations

Pie — chosen by all three students

Tim — chose at least one item from every station

Uta — also chose at least one item from every station

Yuen — chose no salad

Use the matrix to test each quantified claim.

Cafeteria claims A and B

A. There is an item chosen by every student.

$$\exists i \; \forall s \;\; \text{Chosen}(s,i)$$

TRUE

Witness: pie.
Uta, Tim, and Yuen all chose it.

One item must work for all students.

B. There is a student who chose every item.

$$\exists s \; \forall i \;\; \text{Chosen}(s,i)$$

FALSE

No student chose all nine items.
Each possible student fails on several items.

A universal inner condition needs complete coverage.

Cafeteria claims C and D

C. Some student chose an item from every station.

$$\exists s\ \forall z\ \exists i\ [\text{ItemIn}(i,z) \wedge \text{Chosen}(s,i)]$$

TRUE

Witnesses: Uta or Tim.
For each station z, a suitable chosen item i can depend on z.

D. Every student chose an item from every station.

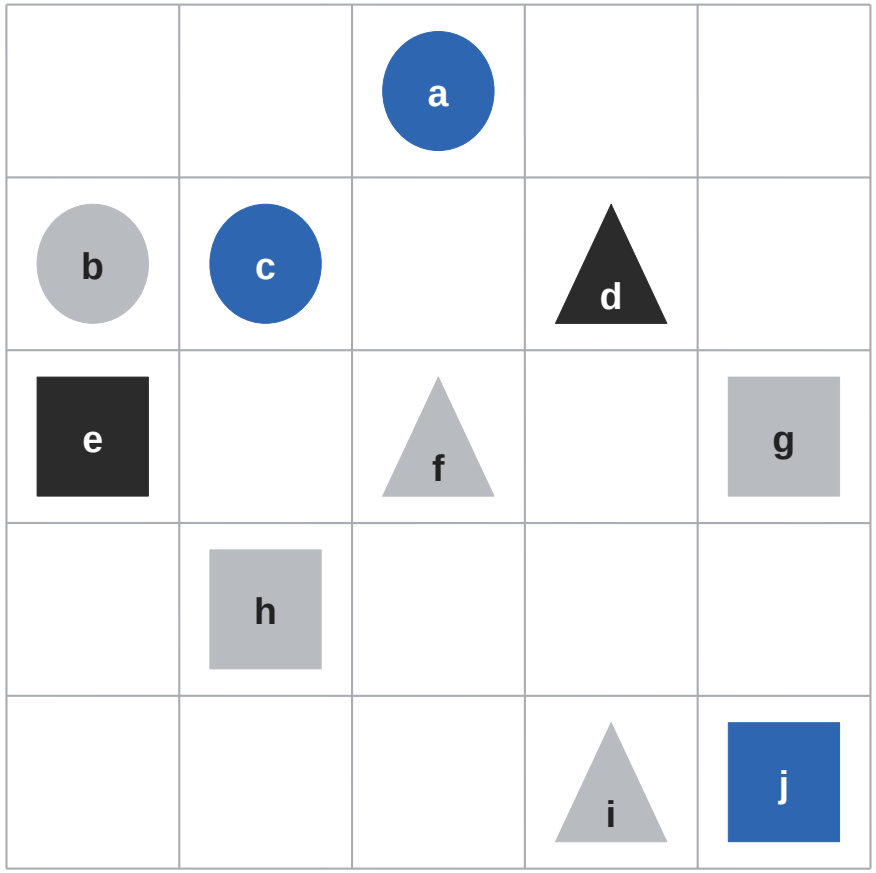
$$\forall s\ \forall z\ \exists i\ [\text{ItemIn}(i,z) \wedge \text{Chosen}(s,i)]$$

FALSE

Counterexample: student Yuen and the salad station.
No salad item i satisfies Chosen(Yuen,i).

Nested quantifiers encode a sequence of dependent choices.

Tarski world: a finite geometric model



Common domain

All ten objects a-j in the grid.

Unary predicates

Circle(x), Square(x), Triangle(x)
Blue(x), Gray(x), Black(x)

Binary predicates

Above(x,y) RightOf(x,y) SameColor(x,y)

Read a formula by tracing its domain restrictions, quantifier order, and spatial relation.

Formalizing statements in Tarski world

English statement	Formal statement	Truth
Every circle is above f.	$\forall x(\text{Circle}(x) \rightarrow \text{Above}(x,f))$	True
There is a black square.	$\exists x(\text{Square}(x) \wedge \text{Black}(x))$	True
Every circle has a square of the same color.	$\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x,y)))$	True
There is a square to the right of every triangle.	$\exists x(\text{Square}(x) \wedge \forall y(\text{Triangle}(y) \rightarrow \text{RightOf}(x,y)))$	True

Check the grammar of the domain restriction

$\forall x:\ \text{Type}(x) \rightarrow \text{property}$

$\exists x:\ \text{Type}(x) \wedge \text{property}$

Restricted domains: implication versus conjunction

When the common domain is larger than the intended subdomain, restricted quantifiers are abbreviations.

Restricted form	Unrestricted-domain form	Why
$\forall x \in D \ P(x)$	$\forall x(D(x) \rightarrow P(x))$	Only members of D are required to satisfy P.
$\exists x \in D \ P(x)$	$\exists x(D(x) \wedge P(x))$	The witness must be both in D and satisfy P.

Universal example $\forall x(Student(x) \rightarrow Passed(x))$ Every student passed.

Existential example $\exists x(Student(x) \wedge Passed(x))$ Some student passed.

Formalization example: reciprocals

Definition: b is a reciprocal of a exactly when ab = 1.

Every nonzero real number has a reciprocal.

$$\forall u \in \mathbb{R}\{0\}\ \exists v \in \mathbb{R}\ \ (uv = 1)$$

Given u ≠ 0, choose v = 1/u.

There is a real number with no reciprocal.

$$\exists c \in \mathbb{R}\ \forall d \in \mathbb{R}\ \ (cd \neq 1)$$

Witness: c = 0, because 0·d = 0 for every real d.

Boundary case The restriction u ≠ 0 is essential. Without it, the first statement is false.

Two order-sensitive number statements

There is a smallest positive integer.

$$\exists m \in \mathbb{Z}^+ \; \forall n \in \mathbb{Z}^+ \; (m \leq n)$$

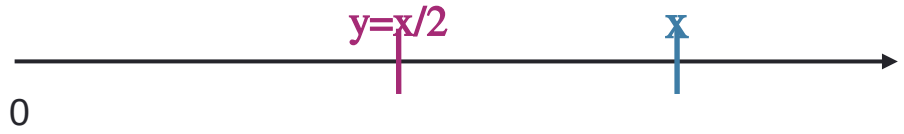
TRUE. The common witness is m = 1.



There is no smallest positive real number.

$$\forall x \in \mathbb{R}^+ \; \exists y \in \mathbb{R}^+ \; (y < x)$$

TRUE. After x is chosen, choose y = x/2.



Same topic, different witness structure: one fixed minimum vs. a smaller value chosen after x.

Negating multiple quantifiers: a mechanical method

Quantifier rules

$$\neg\forall \equiv \exists\neg$$

$$\neg\exists \equiv \forall\neg$$

Move the negation inward one quantifier at a time.

General pattern

$$\neg(Q_1x_1\ Q_2x_2 \cdots Q_nx_n\ P)$$

Flip each quantifier and negate the final predicate.

Three-quantifier example

$$\neg\forall x\exists y\forall z\ P(x,y,z)$$

$$\equiv \exists x\forall y\exists z\ \neg P(x,y,z)$$

Sanity check: the negation must describe exactly how the original statement can fail.

Worked negations with Loves(x,y)

Original claim	$\forall x \exists y\ Loves(x,y)$	Everyone loves someone.
Negation	$\exists x \forall y\ \neg Loves(x,y)$	**There is a person who loves nobody.**

Original claim	$\exists x \forall y\ Loves(x,y)$	Someone loves everyone.
Negation	$\forall x \exists y\ \neg Loves(x,y)$	**Every person fails to love at least one person.**

Not all ≠ none. Preserve the full quantifier structure when verbalizing a negation.

Negation with domains and connectives

Original statement: every circle has a square of the same color.

$$\forall x(\text{Circle}(x) \rightarrow \exists y(\text{Square}(y) \wedge \text{SameColor}(x,y)))$$

1. Flip ∀ $\exists x\, \neg(\text{Circle}(x) \rightarrow \exists y(...))$

2. Negate → $\exists x(\text{Circle}(x) \wedge \neg\exists y(...))$

3. Flip ∃ $\exists x(\text{Circle}(x) \wedge \forall y\, \neg(\text{Square}(y) \wedge \text{SameColor}(x,y)))$

4. Apply De Morgan $\exists x(\text{Circle}(x) \wedge \forall y(\neg\text{Square}(y) \vee \neg\text{SameColor}(x,y)))$

English negation: there is a circle that has no square of the same color.

Common errors - and how to repair them

Swapping mixed quantifiers

$$\forall x \exists y\, P \;\Rightarrow\; \exists y \forall x\, P$$

Ask whether one common witness exists.

Flipping only the first quantifier

$$\neg \forall x \exists y\, P \;\Rightarrow\; \exists x \exists y \neg P$$

Push ¬ inward through every quantifier.

Using ∧ in a universal restriction

$$\forall x(\text{Student}(x) \wedge \text{Passed}(x))$$

Use Student(x) → Passed(x).

Confusing “not all” with “none”

$$\neg \forall x\, P(x) \;\Rightarrow\; \forall x \neg P(x)$$

Correct: ∃ x¬P(x).

Repair strategy: verbalize the formula, test a tiny model, then formalize again.

Application: access-control policies

Access(u,s): user u can access server s.

Every user can access some server.

$$\forall u \, \exists s \;\; Access(u,s)$$

Different users may have access to different servers.

One server is accessible by every user.

$$\exists s \, \forall u \;\; Access(u,s)$$

This is a much stronger - and potentially risky - policy.

Negation of the first policy

$$\neg \forall u \exists s \; Access(u,s) \;\equiv\; \exists u \forall s \; \neg Access(u,s)$$

There is a user who cannot access any server.

EXTENSION

AI knowledge representation: the Nono scenario

Natural-language knowledge base

The law says it is a crime for an American to sell weapons to hostile nations. Nono is an enemy of America, owns some missiles, and all its missiles were sold to it by Colonel West, who is American.

Predicate dictionary

American(x)
x is American

Weapon(y)
y is a weapon

Sells(x,y,z)
x sells y to z

Hostile(z)
z is hostile

Owns(z,y)
z owns y

Missile(y)
y is a missile

Enemy(z,America)
z is an enemy of America

Criminal(x)
x is a criminal

The first task is not “reasoning”; it is precise formalization with correct scope.

EXTENSION

Nono scenario: formal rules and inference

$$\forall x \forall y \forall z\ [(\text{American}(x) \land \text{Weapon}(y) \land \text{Sells}(x,y,z) \land \text{Hostile}(z)) \rightarrow \text{Criminal}(x)]$$

$$\exists m\ [\text{Owns}(\text{Nono},m) \land \text{Missile}(m)]$$

$$\forall m\ [(\text{Missile}(m) \land \text{Owns}(\text{Nono},m)) \rightarrow \text{Sells}(\text{West},m,\text{Nono})]$$

$$\forall m\ [\text{Missile}(m) \rightarrow \text{Weapon}(m)]$$

$$\forall z\ [\text{Enemy}(z,\text{America}) \rightarrow \text{Hostile}(z)]$$

$$\text{American}(\text{West}) \qquad \text{Enemy}(\text{Nono},\text{America})$$



EXTENSION

Lecture synthesis: one connected reasoning cycle

1. Natural language	2. Formalize	3. Interpret order	4. Test truth	5. Negate / apply
Find quantifier words and possible ambiguity.	Choose domains, predicates, and argument order.	Identify which witnesses may depend on earlier choices.	Search for witnesses or counterexamples.	Flip quantifiers, negate the condition, and use the result.

$\forall x \exists y$: y may depend on x | $\exists y \forall x$: one y must work for all x

Key sentence: quantifier order determines dependency.

LEARNING APPENDIX

Revision and assessment

A

Core concepts, connections, worked examples, knowledge checks, and verified solutions.

Review • Practice • Check

Appendix A: core concepts and misconceptions

Multiply quantified statement	contains more than one quantifier.
Dependency	a later existential witness may depend on earlier universal choices.
Common witness	an existential value chosen before a universal challenge must work for all cases.
Negation	flip every quantifier and negate the final condition.
Restricted domain	∀ uses →; ∃ uses ∧.

Misconceptions to avoid

- ✗ Mixed quantifiers can always be swapped.
- ✗ One witness is required for every ∀x∃y claim.
- ✗ “Not all” means “none.”
- ✗ Negating the predicate alone is enough.
- ✗ Variable position in P(x,y) does not matter.

Best habit: paraphrase every formula in precise English before manipulating it.

Appendix B: conceptual connections

Structure	Operational reading	Typical proof / disproof move
$\forall x \exists y\ P(x,y)$	An adversary chooses x; then choose y.	Prove: give a witness rule y=f(x). Disprove: find an x for which no y works.
$\exists y \forall x\ P(x,y)$	Choose one y before seeing x.	Prove: give one common witness. Disprove: defeat each candidate y with some x.
$\neg\forall x \exists y\ P$	Some x has no successful y.	Equivalent to $\exists x \forall y \neg P$.
$\neg\exists y \forall x\ P$	Every candidate y fails somewhere.	Equivalent to $\forall y \exists x \neg P$.
$\forall x \in D\ P(x)$	Every member of D has P.	Write $\forall x(D(x) \rightarrow P(x))$.
$\exists x \in D\ P(x)$	Some member of D has P.	Write $\exists x(D(x) \wedge P(x))$.

Appendix C: representative worked examples

Valid ∀∃

$$\forall x \in \mathbb{Z}\ \exists y \in \mathbb{Z}\ (x+y=0)$$

Choose y=-x.

Invalid ∃∀

$$\exists y \in \mathbb{Z}\ \forall x \in \mathbb{Z}\ (x+y=0)$$

No single y cancels all x.

Boundary case

$$\forall x \in \mathbb{R}^{+}\ \exists y \in \mathbb{R}^{+}\ (y<x)$$

Choose y=x/2; hence no smallest positive real.

Application

$$\neg \forall u \exists s\ \mathrm{Access}(u,s)$$

Equivalent to ∃u∀s¬Access(u,s).

Each example exercises a distinct case: dependent witness, common witness failure, dense-order boundary, and policy negation.

Appendix D: knowledge check 1

1. Multiple choice **Which formula means “one server is accessible by every user”?**

A. $\forall u \exists s\ \text{Access}(u,s)$

B. $\exists s \forall u\ \text{Access}(u,s)$

C. $\forall s \exists u\ \text{Access}(u,s)$

D. $\exists u \forall s\ \text{Access}(u,s)$

---

2. Justified true / false **Swapping ∀ and ∃ never changes the truth value of a statement.**

3. Justified true / false $\forall x \forall y\ P(x,y) \equiv \forall y \forall x\ P(x,y)$

For each true/false item, give one sentence of justification.

Appendix D: knowledge check 2

4. Formalize **Every student owns at least one laptop.**

Use a common domain and the predicates Student(x), Laptop(y), and Owns(x,y).

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5. Negate $\forall x \exists y\, R(x,y)$

Give both the symbolic negation and an English reading.

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6. Error detection **A student translates “Some student passed” as:**

$$\forall x(\text{Student}(x) \rightarrow \text{Passed}(x))$$

Identify the error and write the correct formula.

Appendix D: knowledge check 3

7. Construct a countermodel

On the domain {1,2}, define a relation R such that

$$\forall x \exists y\ R(x,y) \text{ is true, but } \exists y \forall x\ R(x,y) \text{ is false.}$$

Give R as a set of ordered pairs and justify both truth values.

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8. Security-policy analysis

Negate and interpret: $\forall u \exists s\ Access(u,s)$

Then explain whether the negation describes one user, every user, one server, or every server.

Show your reasoning; a correct formula without interpretation is incomplete.

Appendix E: answer key 1

1 **B. ∃ s∀ u Access(u,s).**

The server is chosen once and must work for every user. A allows different servers; C and D reverse the roles.

2 **False.**

Mixed quantifiers generally cannot be swapped. Example: ∀x∃ y(x+y=0) is true over $\mathbb{Z}$, but ∃ y∀ x(x+y=0) is false.

3 **True.**

Adjacent universal quantifiers commute: both formulas require P(x,y) for every ordered pair (x,y).

4 $\forall x(\text{Student}(x) \rightarrow \exists y(\text{Laptop}(y) \wedge \text{Owns}(x,y)))$.

The universal restriction uses implication; the existential laptop witness uses conjunction.

Appendix E: answer key 2

5 $\exists x \forall y \neg R(x,y)$.

English: there is an x related to no y. Common error: flipping ∀ but leaving ∃ unchanged.

6 $\exists x(\text{Student}(x) \wedge \text{Passed}(x))$.

The original formula says every student passed. “Some” requires ∃, and the witness must satisfy both predicates.

7 $R = \{(1,1),(2,2)\}$.

Each x has a witness y=x, so ∀x∃y is true. No single y is related to both 1 and 2, so ∃y∀x is false.

8 $\exists u \forall s \neg \text{Access}(u,s)$.

There is one user who cannot access any server. The outer existential identifies a particular failing user.

Appendix F: optional enrichment

Calculus: limits

A sequence limit contains alternating choices: $\forall \varepsilon > 0\ \exists N\ \forall n > N,\ |a_n - L| < \varepsilon$.

N may depend on ε, but once chosen it must work for all later n.

Logic programming

Prolog represents facts and rules, then uses search and inference to answer queries.

FOL notation makes such reasoning mechanically checkable.

Database constraints

“Every order has a customer” has form ∀o∃c.

“Some administrator can access every table” has form ∃a∀t.

The order changes the constraint.

Quantifier alternation appears whenever choices, dependencies, or guarantees occur in sequence.

References and course resources

Primary mathematical reference

Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., Section 3.3.

Original course materials

Prof. Mustafa Jarrar - Discrete Mathematics course materials

Updated course resources

COMP233 GitHub - BZU Courses

Acknowledgment Local lecture continuity also draws on materials associated with Dr. Radi Jarrar. The reconstruction corrects notation and scope while preserving the original teaching intent.